

HAMZA NKHUMBWA

GIS & Data Analyst | Earth Observation Specialist | Licensed UAV Pilot

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PROFESSIONAL PROFILE

GIS and data analyst with an Earth Observation background (BSc, MUST) and a licensed drone pilot, with 3+ years of hands-on experience turning drone imagery, satellite data, field surveys and administrative records into maps, dashboards and decisions. Delivered work for Illovo Sugar estates, the National Water Resources Authority, Lilongwe City Council and the national SBAPP land cover mapping programme. Strong in Python, R, SQL, Power BI, Google Earth Engine and ArcGIS Pro/QGIS. Seeking a GIS, data analysis, remote sensing or UAV role where data drives real-world decisions.

CORE SKILLS

Data Analysis & Visualisation: Python, R, SQL, SPSS, Power BI, interactive dashboards, statistical analysis, accuracy assessment, database design

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online, QGIS, Google Earth Engine, ENVI, Digital Earth Africa

UAV & Surveying: Licensed remote pilot, orthomosaic and 3D mapping, multi-sensor imagery processing, GNSS and topographic surveys

Analysis & Modelling: Land cover change detection, machine learning prediction, accuracy assessment, water basin delineation, temporal analysis

Programming & Databases: Python, JavaScript, R, SQL, PostgreSQL/PostGIS, MySQL

Dashboards & Web GIS: Power BI, HTML/CSS, React, GitHub, Vercel

Professional: Stakeholder engagement, technical report writing, project leadership, teamwork

Languages: English, Chichewa, Setswana

PROFESSIONAL EXPERIENCE

Drone Pilot, Monitoring & Surveillance

Present

World Wide Joint Discoveries (subcontractor to Alumenda Illovo Sugar Estates)

- Flew UAV monitoring and surveillance missions across Illovo estates, capturing high-resolution imagery for environmental monitoring, estate planning and operational assessment.
- Processed imagery into orthomosaics and map products covering 800 hectares and built GIS workflows for mapping, analysis and reporting.
- Developed a spatial decision-making dashboard for Alumenda Estate to support sugarcane agriculture sustainability.
- Operated as security drone pilot, flying continuous patrols around the estate.

GIS Intern (Volunteer)

June 2025 – Feb 2026

National Water Resources Authority (NWRA)

- Built an automated, real-time interactive dashboard for hydrogeological decision-making.
- Designed a borehole database for the groundwater section, mapping spatial distribution of sources to support planning (4,650 boreholes).
- Produced land use and land cover maps using JavaScript in Google Earth Engine, and delineated water basins from gauging station coordinates.
- Inspected surface and groundwater abstractors with GNSS receivers and verified compliance of abstraction permits and licences.

GIS & Mapping Intern

1 year

Lilongwe City Council, Town Planning Office

- Digitised land parcels in more than 3 areas for regularisation and official site plans.
- Produced high-quality and high-level planning maps in ArcGIS Pro and QGIS.
- Worked with the Chief Town Planning Officer on the UN Urban Resilience Plan and carried out topographic surveys.

PROJECTS & RESEARCH

Dzalanyama Forest Change Analysis & Future Prediction

4 months

Personal research project, manuscript available online

- Analysed land cover change using Google Earth Engine, R, ArcGIS and ArcGIS Pro, and predicted future scenarios with machine learning.
 - Carried out accuracy assessment (overall and user accuracy) for the manuscript.
- Manuscript: [Read the manuscript](#)

Spatial Biodiversity Assessments, Prioritization & Planning (SBAPP)

1 week

National land cover mapping

- Contributed to Malawi's official land cover maps in Google Earth Engine, working with the Departments of Surveys, Forestry, Environmental Affairs, Land Resources, Physical Planning and the Malawi Environmental Protection Authority.

Salima Pre-Disaster Preparedness Mapping

2024

African Drone and Data Academy Emergency Unit, supported by UNICEF

- Contributed to participatory baseline mapping across more than 305 hectares using KoboCollect, GPS Essential and a DJI Mavic 3E Enterprise drone, to support disaster preparedness planning.

Drone-Based Soil Fertility Assessment System

3 weeks

Agriculture

- Built a drone-based system for assessing soil fertility to support agricultural decisions.

Topographic Survey, Golomoti

2024, 3 days

Field survey

- Collected coordinates with GNSS/GPS, surveyed ground features and boundaries, and produced satellite-based site maps with accurate hectare calculations.

EDUCATION

BSc Geo-Information Systems & Earth Observation Science (Credit)

Feb 2021 – May 2025

Malawi University of Science and Technology (MUST)

Malawi School Certificate of Education, LL TTC Open Secondary School (2019)

LICENCES & CERTIFICATIONS

- **Remote Pilot Licence** and medical fitness certificate, Malawi Civil Aviation Authority
- **Certificate in Drone Operation, Level 2**, African Drone and Data Academy (MUST & Virginia Tech), Oct 2025
- **Certificate in Drone Operation, Level 1**, African Drone and Data Academy, Dec 2024 – Jan 2025
- **Spatial Analysis: New Frontiers and Making an Impact with Modern Geo Apps (ArcGIS Pro & Online)**, ESRI Academy, 2024
- **Digital Earth Africa: Water Management and Sandbox Introduction**
- **Software Engineering**, ALX, 2024

AWARDS & RECOGNITION

- FDH–MUST Graduate Innovative Ideas Award, 2026
- Small Grants Award for research, OpenStreetMap (OSM) Malawi
- Ranked Top 1 and Top 2 UAV Maneuver Operator, 2025

LEADERSHIP & COMMUNITY

- **Interim President and Academic Director**, Student Representative Council, MUST
- **Drone Technology President**, Cohort16-ADDA Class :lead pre- and post-flight safety checks, mission planning and supervision of UAV operations
- **Organising member**, State of Map Malawi (LUANAR)
- Contributed to forest and sanctuary restoration for the Community Approach to Waterbird Resources Management, Chia Lagoon

REFEREES

Mr. Chikondi Chiuimbia, CEO, World Wide Joint Discoverers
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Mr. Mike Myaba, Senior Hydrologist, NWRA
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